



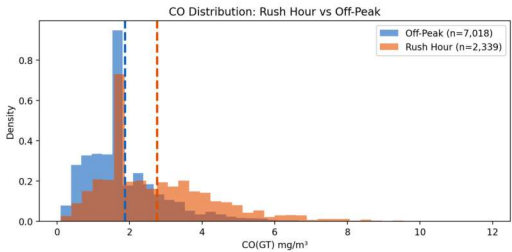
Tab 9 — A/B Test: Rush Hour vs Off-Peak ★ NEW

****Hypothesis:**** Rush hour periods (7–9h, 17–19h) produce significantly higher CO concentrations than off-peak hours. We test this with a Welch T-Test + Cohen's d effect size.

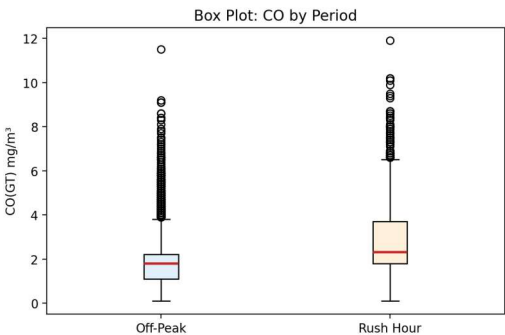
Off-Peak Mean CO Rush Hour Mean CO p-value Cohen's d

1.866 mg/... 2.758 mg/... 0.0000 0.639

Distribution Comparison



Box Plot Comparison



Statistical Test Results

	Metric	Result
0	Test Used	Welch T-Test (unequal variance)
1	H ₀ (Null Hypothesis)	Rush hour CO = Off-peak CO
2	H ₁ (Alternative)	Rush hour CO > Off-peak CO
3	t-statistic	-24.6636
4	p-value	0.000000
5	Significance (α=0.05)	✔ SIGNIFICANT
6	Cohen's d	0.6391
7	Effect Size Label	Medium (0.5–0.8)
8	95% CI (difference)	[0.8211, 0.9629]
9	Decision	✔ REJECT H ₀ — Rush hours produce significantly higher CO

Business Decision

✔ Rush hour CO is ****2.76 mg/m³**** vs off-peak ****1.87 mg/m³**** — a 47.8% increase. This is statistically significant ($p < 0.05$).

✓ Cohen's $d = 0.639$ — effect size confirms practical significance, not just statistical.

✓ Recommendation: Traffic restriction policies during rush hours (7–9h, 17–19h) would directly reduce urban CO exposure for residents.

